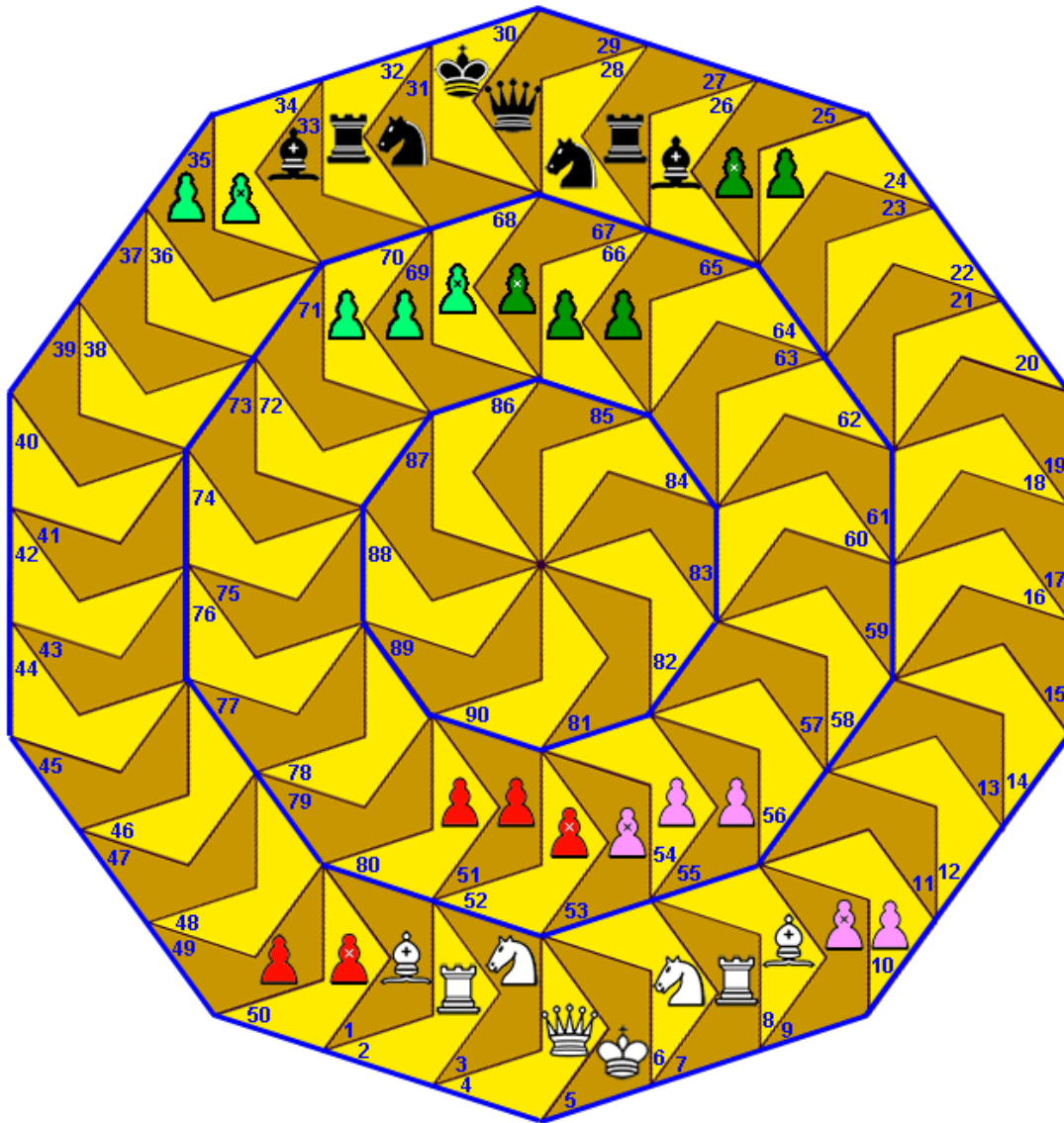


Impossible Chess Made Possible

PENTACHESS

McMaster Capstone Project



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Project Description

Supervisor

Name: Dr. Paul Rapoport

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Organization: McMaster University

Purpose

Chess is a timeless strategy game that has existed for centuries. It enables individuals to engage in a stimulating mental challenge while also enjoying themselves. The inherent social component in this two-player game provides the foundation for strong communities of players. Moreover, we propose to provide a fascinating twist on the standard version of chess that players are used to, called Pentachess. This playable version of Pentachess will allow experienced and novice chess players an amazing opportunity to reignite their passion for the game, encouraging them to spend their free time in an engaging and healthy way. We hope this initiative will inspire individuals to think creatively and strengthen bonds within communities with the joy of chess. The stakeholders involved in this project will be chess players (both novice and experienced), the project supervisor (Dr. Paul Rapoport), and those working on the project directly such as designers, programmers, and testers.

Details

This full-stack web application will allow users to play 1-on-1 games online in real-time. The game features 36 pieces, with each player allocated 18. Each piece type moves and interacts based on specific rules, which will be provided to each player. When it is either player's turn, they can view all possible movements of the piece they click on. Once a piece is selected, all pentagons where the piece can legally move are highlighted. The user must drag the selected piece on one highlighted spot to confirm their move. Moreover, the user can view all their captured pieces and the enemy's captured pieces through an interface beside the chess board. After each move, the game will check for conditions to determine if there's a checkmate or stalemate, announcing the winner or declaring a draw as needed. Additionally, there will be an implemented timer which can be set based on the initial game setup. This timer will ensure there are no idle players. After each move, an additional 5 seconds will be added to the player's timer, in accordance with professional chess rules.

Plan

As mentioned previously, we will be building a full-stack web application. With this in mind, our project will consist of programming languages best suited for web applications. These consist of HTML, CSS, JavaScript (React), and/or Kotlin. For the backend, we will need to use frameworks and libraries for creating REST APIs (node.js, express.js), managing web socket connections (socket.io), and ORMs (Prisma) for interfacing with the database which will either be SQL (PostgreSQL, MySQL) or NoSQL (MongoDB) based. For project management tools, we will use features built-in to GitHub, such as issues, pull requests, README docs, and projects. Looking at the data aspect, the only necessary data is for the design of all the pieces, the design of the board, and the rules of the game/pieces. Most of this data has already been kindly provided by our supervisor, Dr. Paul Rapoport. The only data left to acquire is the design of all the pieces (in image format). We can create designs for these ourselves if Dr. Paul does not already have access to these images.